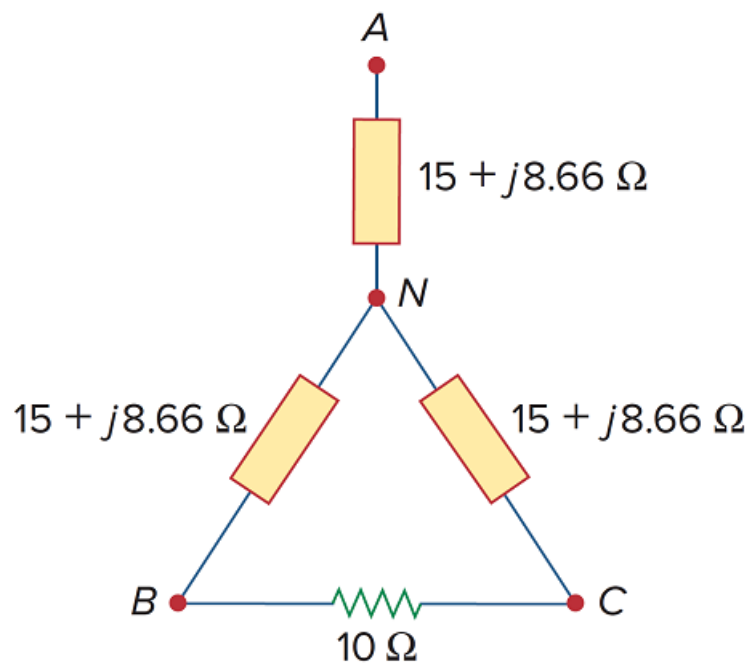


Problem

A three-phase supply, with the line-to-line voltage of 240 V rms, has the unbalanced load as shown in Fig. 12.62. Find the line currents and the total complex power delivered to the load.

Figure 12.62



Step-by-step solution

Step 1 of 11

Refer to Figure 12.62 from the textbook.

Consider the line voltage as:

$$V_{AB} = 240 \angle 0^\circ \text{ V}$$

Assume the phase sequence as ABC .

Since the line voltages have a phase difference of 120° between them.

Thus,

$$V_{BC} = 240 \angle -120^\circ$$

$$V_{CA} = 240 \angle 120^\circ \text{ V}$$

Step 2 of 11

Since the phase voltages lags their respective line voltages by an angle of 30° .

Thus,

The phase voltage, V_{AN} for the star connected system is,

$$V_{AN} = \frac{V_{AB}}{\sqrt{3}} \angle -30^\circ \dots\dots (1)$$

Similarly,

Write the expressions for the phase voltages, V_{BN} and V_{CN} .

$$V_{BN} = \frac{V_{BC}}{\sqrt{3}} \angle -30^\circ \dots\dots (2)$$

$$V_{CN} = \frac{V_{CA}}{\sqrt{3}} \angle -30^\circ \dots\dots (3)$$

Step 3 of 11

Substitute $240\angle 0^\circ$ V for V_{AB} in equation (1).

$$\begin{aligned} V_{AN} &= \left(\frac{240\angle 0^\circ}{\sqrt{3}} \right) \angle -30^\circ \\ &= 138.56\angle -30^\circ \text{ V} \end{aligned}$$

Substitute $240\angle -120^\circ$ V for V_{AB} in equation (2).

$$\begin{aligned} V_{BN} &= \left(\frac{240\angle -120^\circ}{\sqrt{3}} \right) \angle -30^\circ \\ &= 138.56\angle -150^\circ \text{ V} \end{aligned}$$

Substitute $240\angle 120^\circ$ V for V_{AB} in equation (2).

$$\begin{aligned} V_{CN} &= \left(\frac{240\angle 120^\circ}{\sqrt{3}} \right) \angle -30^\circ \\ &= 138.56\angle 90^\circ \text{ V} \end{aligned}$$

Step 4 of 11

Write the value of impedance, Z_Y in polar form.

$$\begin{aligned} Z_Y &= 15 + 8.66j \Omega \\ &= 17.204\angle 30^\circ \Omega \end{aligned}$$

Write the expression for the phase current, I_{AN} .

$$I_{AN} = \frac{V_{AN}}{Z_Y}$$

Substitute $138.56\angle -30^\circ$ V for V_{AN} and $17.204\angle 30^\circ \Omega$ for Z_Y .

$$\begin{aligned} I_{AN} &= \frac{138.56\angle -30^\circ}{17.204\angle 30^\circ} \\ &= 8\angle -60^\circ \text{ A} \end{aligned}$$

Since for a star-connected system, the line and phase current are the same.

Thus,

The line current, I_a is,

$$\begin{aligned} I_a &= I_{AN} \\ &= 8\angle -60^\circ \text{ A} \end{aligned}$$

Therefore, the line current, I_a is $\boxed{8\angle -60^\circ \text{ A}}$.

Step 5 of 11

Write the expression for the phase current, I_{BN} .

$$I_{BN} = \frac{V_{BN}}{Z_Y}$$

Substitute $138.56\angle-150^\circ$ V for V_{BN} and $17.204\angle30^\circ \Omega$ for Z_Y .

$$\begin{aligned} I_{BN} &= \frac{138.56\angle-150^\circ}{17.204\angle30^\circ} \\ &= 8\angle180^\circ \text{ A} \end{aligned}$$

Write the expression for the phase current, I_{CN} .

$$I_{CN} = \frac{V_{CN}}{Z_Y}$$

Substitute $138.56\angle90^\circ$ V for V_{CN} and $17.204\angle30^\circ \Omega$ for Z_Y .

$$\begin{aligned} I_{CN} &= \frac{138.56\angle90^\circ}{17.204\angle30^\circ} \\ &= 8\angle60^\circ \text{ A} \end{aligned}$$

Since a resistance of 10Ω is connected between B and C , the load is unbalanced and hence, the line currents, I_b and I_c are not equal to their phase voltages, I_{BN} and I_{CN} .

Step 6 of 11

Assume the current through the resistor be I_{BC} in the direction from C to B .

Apply Kirchhoff's Voltage law in the loop $cnbBC$.

$$10I_{BC} = V_{BC}$$

Step 7 of 11

Substitute $240\angle-120^\circ$ V for V_{BC} .

$$\begin{aligned} 10I_{BC} &= 240\angle-120^\circ \\ I_{BC} &= 24\angle-120^\circ \text{ A} \end{aligned}$$

Step 8 of 11

Apply Kirchhoff's Voltage law in the loop BNC .

$$I_{BC}R + Z_Y I_{BN} - Z_Y I_{CN} = 0$$

Substitute $17.204\angle30^\circ \Omega$ for Z_Y , 10Ω for R , $8\angle180^\circ$ A for I_{BN} and $24\angle-120^\circ$ A for I_{BC} .

$$(24\angle-120^\circ)(10) + (17.204\angle30^\circ)(8\angle180^\circ) - (17.204\angle30^\circ)(I_{CN}) = 0$$

$$I_{CN} = \frac{(17.204\angle30^\circ)(8\angle180^\circ) + 240\angle-120^\circ}{17.204\angle30^\circ}$$

$$I_{CN} = \frac{-239.193 - j276.662}{17.204\angle30^\circ}$$

$$I_{CN} = -20.1 - j6.975 \text{ A}$$

$$I_{CN} = 21.26\angle-160.84^\circ \text{ A}$$

Step 9 of 11

Apply Kirchhoff's current Law at node B .

$$I_b - I_{BN} + I_{BC} = 0$$

$$I_b = I_{BN} - I_{BC}$$

Substitute $24\angle -120^\circ$ A for I_{BC} and $8\angle 180^\circ$ A for I_{BN} .

$$\begin{aligned} I_b &= 8\angle 180^\circ - 24\angle -120^\circ \\ &= 21.17\angle 79.1^\circ \text{ A} \end{aligned}$$

Therefore, the line current, I_b is $21.17\angle 79.1^\circ$ A.

Step 10 of 11

Apply Kirchhoff's current Law at node C .

$$I_c - I_{CN} + I_{BC} = 0$$

$$I_c = I_{CN} - I_{BC}$$

Substitute $24\angle -120^\circ$ A for I_{BC} and $21.25\angle -160.84^\circ$ A for I_{CN} .

$$\begin{aligned} I_c &= 21.25\angle -160.84^\circ - 24\angle -120^\circ \\ &= 16\angle -120.31^\circ \text{ A} \end{aligned}$$

Therefore, the line current, I_c is $16\angle 120.31^\circ$ A.

Step 11 of 11

Calculate the complex power delivered to the load, S .

$$S = S_A + S_B + S_C$$

Calculate S_A .

$$S_A = V_{AN} I_{AN}^*$$

Substitute $138.56\angle -30^\circ$ V for V_{AN} and $8\angle -60^\circ$ A for I_{AN} .

$$\begin{aligned} S_A &= (138.56\angle -30^\circ)(8\angle -60^\circ)^* \\ &= (138.56\angle -30^\circ)(8\angle 60^\circ) \\ &= 960 + j554.24 \text{ VA} \end{aligned}$$

Calculate S_B .

$$S_B = V_{BN} I_{BN}^*$$

Substitute $138.56\angle -150^\circ$ V for V_{BN} and $8\angle -180^\circ$ A for I_{BN} .

$$\begin{aligned} S_B &= (138.56\angle -150^\circ)(8\angle 180^\circ)^* \\ &= (138.56\angle -150^\circ)(8\angle -180^\circ) \\ &= 960 + j554.24 \text{ VA} \end{aligned}$$

Calculate S_C .

$$\begin{aligned} S_C &= |I_{CN}|^2 Z_C \\ &= (21.26)^2 (17.204\angle 30^\circ) \\ &= 6734.21 + j3888 \text{ VA} \end{aligned}$$

Substitute $960 + j554.24$ VA for S_A , $960 + j554.24$ VA for S_B and $6734.21 + j3888$ VA for S_C .

$$\begin{aligned} S &= 960 + j554.24 + 960 + j554.24 + 6734.21 + j3888 \\ &= 8654 + j4996.48 \text{ VA} \end{aligned}$$

Therefore, the complex power, S is $8654 + j4996.48$ VA.

